



www.feitengacp.com

FEITENG[®] DIPPER

Building Grade ACP
Installation System



ZHANGJIAGANG FEITENG NEW COMPOSITE MATERIALS CO., LTD.
FEITENG COMPOSITES (EUROPE) B.V.

Add: Industriestraat 4,
5804 CK Venray, The Netherlands

Phone: +31- 850 161 962
E-mail: info@feitengacp.eu
www.feitengacp.com

The images and text in this brochure are the property of Zhangjiagang Feiteng Composite New Materials Co., Ltd. and may not be reproduced without permission. Any potential changes to the data in this brochure will not be separately notified.

FEITENG

Contents

FEITENG® DIPPER SERIES	
Introduction	01
Attestation	03
Characteristics	05
FFEITENG® A2	07
FFEITENG® FR	09
Colors & Finishes	11
Product Range	13
Storage & Handling	15
Cleaning & Maintenance	16
Processing Method	17

FEITENG® Fastfix	
Metal Composite Panel Curtain Wall Installation steps	21
Open Curtain Wall-Hanging Code Installation Node	23
Open Curtain Wall-Sub Frame Installation Node	25
Closed Curtain Wall-Sub Frame Installation Node	27
Closed Curtain Wall-Plug in Installation Node	29
Accessories	31

FEITENG® Cases	
Overseas architectural curtain wall projects	32
China architectural curtain wall projects	46

FEITENG® DIPPER SERIES

Introduction

Feiteng New Composite Materials established in 1994, is a one-stop solution provider specializing in research and development, design, manufacturing, processing and sales of metal composite materials.

FEITENG metal composite panels offer exceptional performance, combining lightweight properties with rigidity, superior flatness, durability and ease of maintenance. They are specifically designed for building cladding as well as façade renovation or restoration. FEITENG provides architects and manufacturers with unique visual appeal and exciting design possibilities, offering adaptable solutions for all areas of construction, including residential buildings, public facilities, offices and industrial structures. Additionally, they are ideal for representing commercial identities in spaces such as supermarkets, dealerships, banks and gas stations.

FEITENG metal composite panels feature a unique customization system that can meet the demands of

different architectural requirements, offering options for dimensions, colors, surface finishes, fire-retardant cores and stocked materials.

FEITENG is committed to maximizing the lifespan of our products by adhering to ISO standards. We promise that the entire lifecycle of our products, from raw material to their end-of-life, delivers environmental, social and economic benefits. During the production process of FEITENG metal composite panels, we strive to reduce energy and water consumption, maximize productivity and minimize waste.

Explore FEITENG's deep insights into architectural needs and its innovative thinking, continuously driving progress and development in architectural design.



FEITENG® DIPPER SERIES

Attestation

FEITENG upholds its tradition of craftsmanship and high quality, ensuring the performance of our product by passing 28 internal tests and meeting strict quality standards set by external laboratories. The company has also obtained audits and certifications from most countries and regions worldwide. At FEITENG, we believe that only by continuously striving for excellence can we provide customers with superior products and services. This commitment to excellence has always been our unwavering goal.

EU CE certification	US UL certification	EPD
		
TUV certification	SGS certification	ISO certification
		
ICC certification	Reach certification	RoHS certification
		

EPD

FEITENG has obtained EPD certification based on the ISO 14025 international standard. We provide our customers with relevant information about the potential environmental and architectural impacts of using the product, enhancing transparency in environmental protection. The ultimate goal is to demonstrate that using products with life cycle information can generate more beneficial social and economic impacts for our planet.

In addition, they also contain information on other aspects, such as the release of volatile organic compounds (VOC) during use, as well as detailed information on durability and recyclability. Therefore, this document serves as proof of the product's ecological sustainability in construction.

When you choose to use Feiteng products, you are also choosing to minimize the life cycle impacts, including:

-  More efficient manufacturing with less resource waste
-  A high recycling rate close to 100%
-  Utilization of thermal energy reuse during production
-  Robust durability with a lifespan of up to 50 years
-  Negligible volatile organic compound (VOC) emissions

FEITENG® DIPPER SERIES

Characteristics



FEITENG® DIPPER SERIES is one of our product series which is produced by composing two smooth 0.5mm thick aluminum (or other metal) sheets to a solid non-combustible or fire-retardant core through a high-temperature process. It is specifically designed to allow architects and designers to use ACM as the preferred material for building façades while meeting the fire performance requirements outlined in the International Building Code (IBC).

The product characteristics and advantages of FEITENG®—validated through laboratory testing—include



Wide range of colors & finishes



Durability



Ability to be perforated



Formability

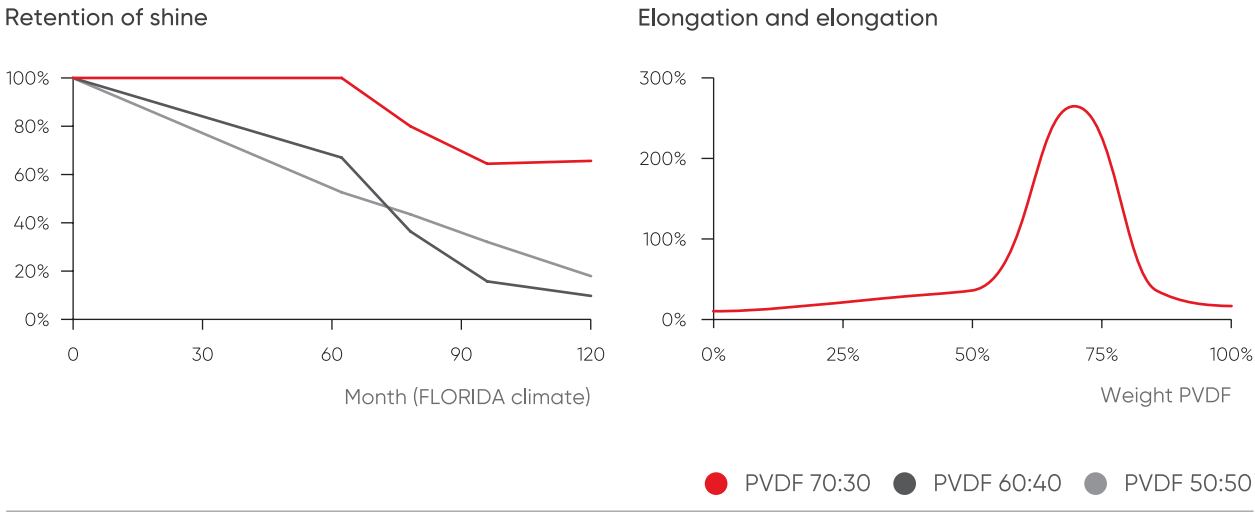


Ease of fabrication



Flatness & Rigidity

FEITENG® offers a variety of colors, gloss levels, textures and finish styles, and we can also provide any color from the RAL, Pantone or NCS color systems. Additionally, FEITENG metal composite panels come in various standard size options, with a maximum width of up to 2000mm.



Based on market demand, FEITENG® DIPPER SERIES offers three different core materials, making the panels either non-combustible or fire-retardant. They are classified (according to China GB8624-2012, Europe EN 13501-1, the U.S. ASTM E84, E119, NFPA 285, and UK BS 476) into FEITENG® DIPPER SERIES A2, FEITENG® DIPPER SEIRES FR, and FEITENG® DIPPER SERIES PE. Each panel is coated with PVDF paint in strict accordance with a 70/30 ratio (containing 70% polyvinylidene fluoride and 30% acrylic resin) to ensure maximum aging resistance and superior UV protection.

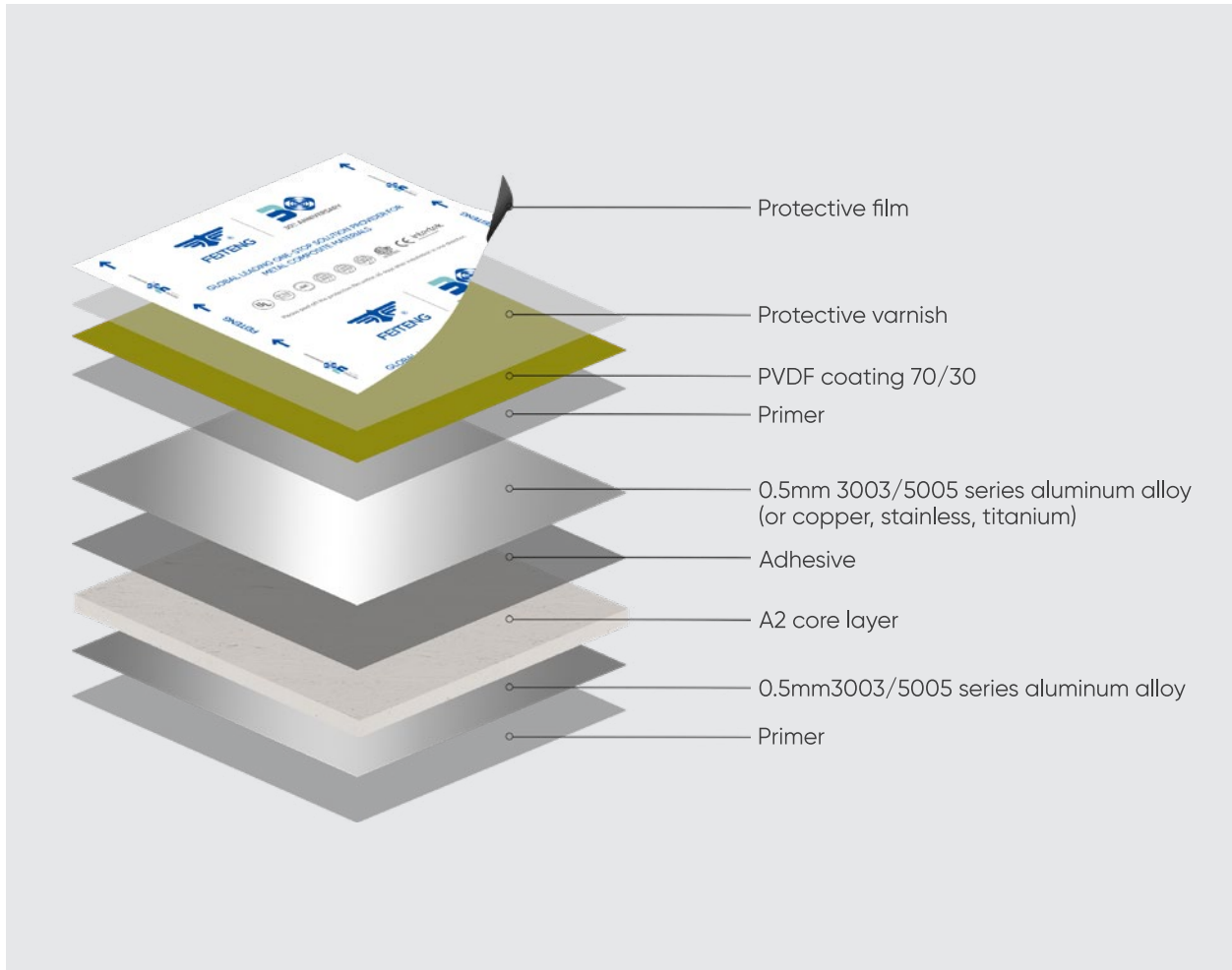
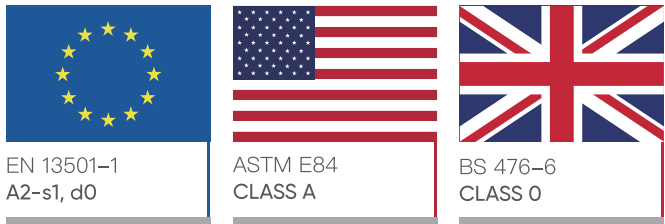
Fire Resistance Classification

Country	Test accord. to...	FEITENG® FR	FEITENG® A2
EU	EN 13501-1	Class B, s1, d0	Class A2, s1, d0
Germany	EN 1187 (method1)/DIN 4102-7	passed	passed
France	NF P 92-501	Class M1	Class M0, non combustible
UK	BS 476-6	Class 0	Class 0
	BS 476-7	Class 1	Class 1
Scandinavia	DS 1085-1		Class A
Switzerland	VKF	Class 5.3	Class 6q.3
Russia	GOST 30244-94	G1	G1
	GOST 30402-95	W1	W1
	GOST 12.1.044-89	D2	D1
	GOST 12.1.044-89	T1	T1
	GOST 31251-03		k0
Australia	AS ISO 9705	Group 1 material	Group 1 material
		SMOGRA 1.385 m2/s2	SMOGRA 0.630 m2/s2
	AS 1530-3 Indices	0 (ignitibility)	0 (ignitibility)
		0 (flame spread)	0 (flame spread)
		0 (heat evolved)	0 (heat evolved)
		0-1 (smoke developed)	0-1 (smoke developed)
	EN 13501-1	B, s1, d0	A2, s1, d0
USA	ASTM-E 84	Class A	Class A
	NFPA285	passed	passed

FEITENG® DIPPER SERIES

FEITENG® DIPPER SERIES A2

FEITENG® DIPPER SERIES A2 is a non-combustible metal composite material that meets the highest fire safety standards for buildings, making it an ideal solution for high-rise structures and high-usage buildings.



Technical Data

Item	units	FEITENG® DIPPER SERIES A2
Thickness	mm	4
Width	mm	1000~2000
Length	mm	2000~8000
Color and surface coating		With monochrome, metal color, magic color, imitation aluminum color, imitation wood grain color and other surface effect series

Special specifications and colors can be negotiated

Performance index

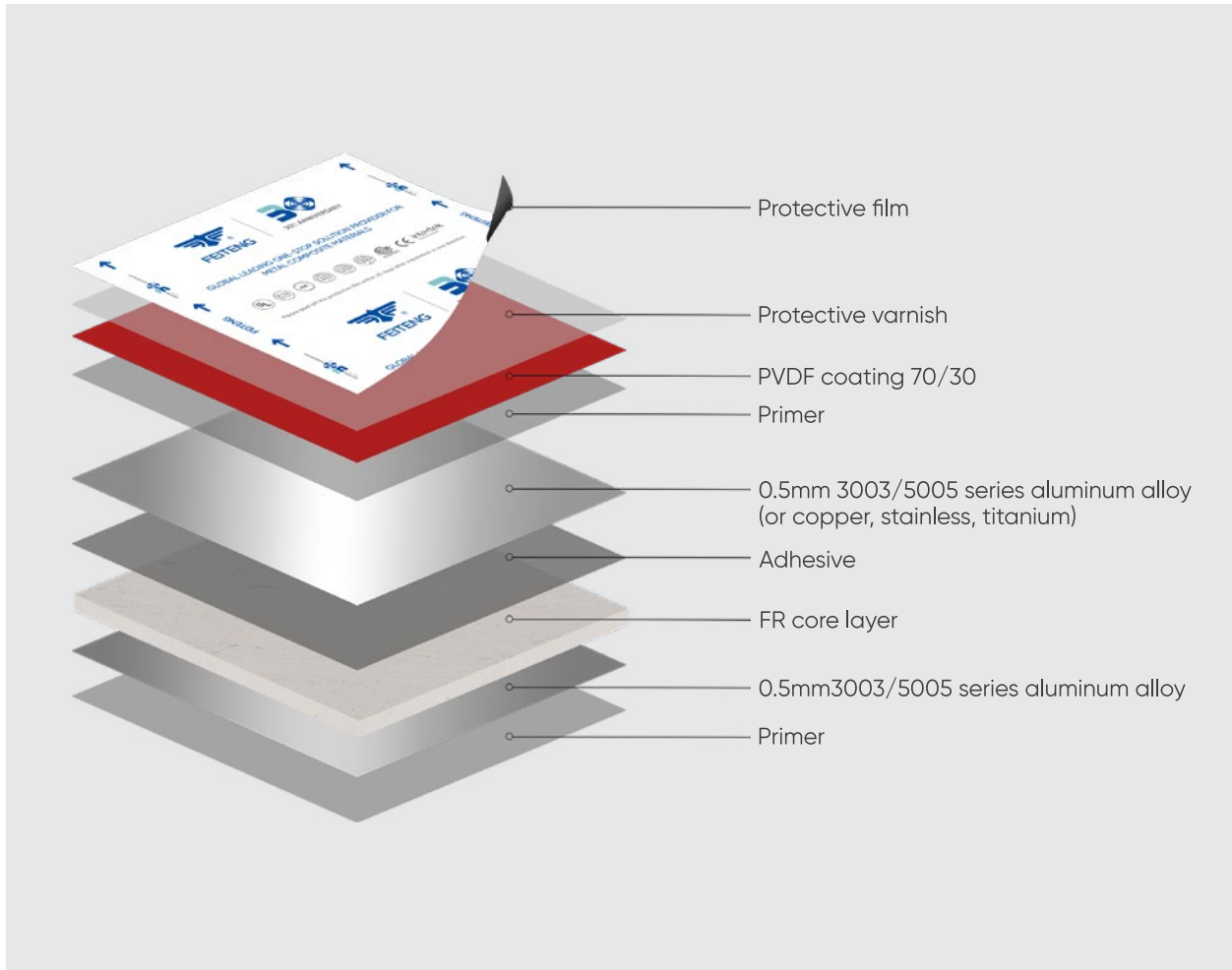
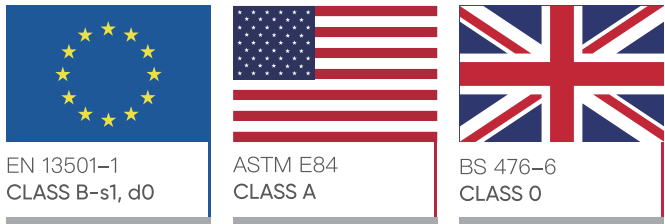
Aluminum thickness on both sides	mm	0.5
Weight	kg/m ²	8.7
Modulus of section	W(cm ³ /m)	1.75
Composite plate stiffness	E.J[kN(cm ² /m)]	2400
Aluminum alloy		3003/5005
Alloy state		H24
Modulus of elasticity	N/mm ²	28000
Tensile strength of aluminum alloy	N/mm ²	R _m ≥130
0.2% yield stress	N/mm ²	R _{p0.2} ≥90
Rate of elongation		A ₅₀ ≥4
Linear coefficient of thermal expansion		2.4mm/m Temperature difference of 100°C
Resistance to temperature difference		-50°C~+80°C
Bending strength	MPa	≥100
Flexural elastic modulus	MPa	≥2.0*10 ⁴
Penetration resistance	kN	≥7
Shear strength	MPa	≥22
Peeling force	[(N·mm)/mm]	Average ≥110, Minimum ≥100
Sound absorption index	α	0.05
Sound insulation index	R _w (dB)	26
Heat transfer coefficient	W/mK	5.8

* The above parameters are measured data of Feiteng laboratory, for reference only, the final data is subject to the actual product

FEITENG® DIPPER SERIES

FEITENG® DIPPER SERIES FR

FEITENG® DIPPER SERIES FR is a commonly used fire-retardant metal composite material that meets the fire safety requirements of most buildings.



Technical Data

Item	units	FEITENG® DIPPER SERIES FR
Thickness	mm	4, 6
Width	mm	800~2050
Length	mm	1500~8000
Color and surface coating		With monochrome, metal color, magic color, imitation aluminum color, imitation wood grain color and other surface effect series

Special specifications and colors can be negotiated

Performance index

Aluminum thickness on both sides	mm	0.5
Weight	kg/m ²	7.5, 10.7
Modulus of section	W(cm ³ /m)	1.75
Composite plate stiffness	E.J[kN(cm ² /m)]	2400, 5900
Aluminum alloy		3003/5005
Alloy state		H24
Modulus of elasticity	N/mm ²	28000
Tensile strength of aluminum alloy	N/mm ²	R _m ≥130
0.2% yield stress	N/mm ²	R _{p0.2} ≥90
Rate of elongation		A ₅₀ ≥4
Linear coefficient of thermal expansion		2.4mm/m Temperature difference of 100°C
Resistance to temperature difference		-50°C~+80°C
Bending strength	MPa	≥100
Flexural elastic modulus	MPa	≥2.0*10 ⁴
Penetration resistance	kN	≥7
Shear strength	MPa	≥22
Peeling force	[(N·mm)/mm]	Average ≥110, Minimum ≥100
Sound absorption index	α	0.05
Sound insulation index	R _w (dB)	26, 27
Heat transfer coefficient	W/mK	5.53, 5.3

* The above parameters are measured data of Feiteng laboratory, for reference only, the final data is subject to the actual product

FEITENG® DIPPER SERIES

Colors & Finishes

Our trend-forward palette represents an ongoing development of colors and finishes curated to foster creativity and ingenuity.

To request samples, please contact: info@feitengacp.com.

Standard Series

01

Color is the soul of architectural art, breathing life into spaces. The ingenious combination of different colors creates contrast and visual focal points. This series of products offers a wide range of options, enabling architects to bring their grand imaginations to life.

Spectra Series

02

To add charm and dynamic elements to any building façade, we skillfully apply the concept of color transitions, creating a vibrant and colorful visual experience on the façade. The seamless color gradients not only enhance the overall visual impact but also inject vitality and energy into the architecture.

Texture Velvet Series

03

With meticulous attention to detail, the designs integrate seamlessly with nature, showcasing diverse textured effects that give each building its own unique "skin".

Diffuse Reflection Series

04

With a reflectivity of $\geq 95\%$, the Diffuse Reflection Series (Patent No.: 202111267860) offers softer and more natural optical effects compared to traditional reflective panels. It creates a more comfortable and inviting visual environment.

Anodized Look Series

05

The FEITENG® Anodized Look Series is a metal composite product manufactured through a special process, offering high recognizability and an anodized-like effect. The metal composite panels undergo multiple steps, including oxidation, pre-roll coating and composite, resulting in corrosion resistance, uniform and stable color and a stronger metallic texture. Additionally, they provide greater cost efficiency.

Copper Stainless Steel Series

06

Both materials exhibit excellent corrosion resistance and weather durability, ensuring the long-term preservation for the exterior appearance of buildings. Copper develops a unique aesthetic through natural oxidation, forming a green patina over time and eventually transitioning into a charming bronze tone. Stainless steel, on the other hand, retains its sleek and polished metallic texture, perfectly suited to the style demands of modern architecture. As architect Frank Gehry once said, "Metal is the material of our time; it transforms architecture into sculpture."

Pure Titanium & Titanium Zinc Series

07

With its exceptional corrosion resistance and aesthetic appeal, titanium is undoubtedly an ideal choice for the architectural industry, especially in areas frequently exposed to marine climates, atmospheric pollution and acid rain.

Brushed Mirror Series

08

This series is an innovative application of architectural materials, which not only enriches the various forms of building exteriors but also introduces cutting-edge technological elements into architectural design. Whether it's the subtle layering of light and shadow or mirror-like façades reflecting the hues of sunrise and sunset, these shimmering creations captivate every passerby with their radiant charm.

Wooden Stone Series

09

One of FEITENG's pro nature products, the TYPE VELVET series and WOODEN STONE series, explores the subtle relationships between humans and nature, as well as between humans and architectural spaces. While pursuing aesthetics and functionality, it also contains deep thinking about modern lifestyles.

Custom Colors



If you need to customize colors for your next project, our technicians will strive to find a match for your color. There is no guarantee for a perfect match because different inks and pigments are used in different industries and applications. Colors achieved through color-matching techniques may not look exactly the same as the colors created through spray application methods, particularly when it comes to metallic colors.

To ensure that we identify your color correctly, we require either:

- A hand sample minimum 1" x 1", OR
- Pantone®, RAL, NCS or paint code reference

Send the color sample along with your name, company name, address, phone number and email, as well as the project name, project location, type of finish and gloss level to:

Feiteng Composites (Europe) B.V.
Industriestraat 4,
5804 CK Venray, The Netherlands

FEITENG® DIPPER SERIES

Product Range



MASS TOLERANCES (STANDARD)

Due to manufacturing, a displacement of the cover sheetssidewise at the panel edges up to 2 mm is possible.
Thickness: ±0.2mm (matt finishstove enamelled anodised)
Lengths: 2000~4000mm; Width: -0 / +6mm
Lengths: 4001~8000mm; Width: -0 / +10mm

FEITENG® A2

Thickness: 4mm (6mm on request)

Width (mm)	1000	1250	1500	1550	1750	2000
Length (mm)	2000~8000	2000~8000	2000~8000	2000~8000	2000~8000	2000~8000
Standard Serie	●	●	●	●	●	●
Wooden Stone Series	●	●	●	●	-	-
Spectra Series	●	●	●	●	●	●
Brushed Mirror Series	●	●	●	-	-	-
Texture Velvet Series	●	●	●	●	●	●
Diffuse Reflection	●	●	●	●	●	●
Copper Stainless Steel	●	●	-	-	-	-
Zinc Series	○	●	-	-	-	-
Anodized Look Series	●	●	●	●	●	●

The delivery time and minimum quantity vary according to size and thicknesses.
Other dimensions are available on request.

○ on request

FEITENG® FR

Thickness: 4mm (6mm on request)

Width (mm)	1000	1250	1500	1550	1750	2000
Length (mm)	2000~8000	2000~8000	2000~8000	2000~8000	2000~8000	2000~8000
Standard Serie	●	●	●	●	●	●
Wooden Stone Series	●	●	●	●	-	-
Spectra Series	●	●	●	●	●	●
Brushed Mirror Series	●	●	●	-	-	-
Texture Velvet Series	●	●	●	●	●	●
Diffuse Reflection	●	●	●	●	●	●
Copper Stainless Steel	●	●	-	-	-	-
Zinc Series	○	●	-	-	-	-
Anodized Look Series	●	●	●	●	●	●

The delivery time and minimum quantity vary according to size and thicknesses.
Other dimensions are available on request.

○ on request

FEITENG® DIPPER SERIES

Storage & Handling



Film Protect

The protective film on FEITENG® metal composite panels plays an essential role during processing, transportation and installation. They prevent the metal surface from accumulating dust and stains. Long time exposure to sunlight can cause the protective film to lose its adhesive properties and become difficult to remove. Therefore, it is crucial to remove the protective film within 45 days after the panels are installed on the wall. Additionally, when installing FEITENG® metal composite panels, ensure that the arrow markings on the protective film are aligned in the same direction and pointing upward.



FEITENG® DIPPER SERIES

Cleaning & Maintenance

1. The surface of FEITENG® metal composite panels requires regular professional cleaning and maintenance. Such cleaning is not only to keep the building looking clean and attractive but also to regularly remove harmful dirt accumulated on the surface, ensuring the long-term quality of the panels.
2. The frequency of cleaning largely depends on the environmental conditions of the project location and the actual level of dirtiness on the panel surface. We recommend cleaning the exterior walls of buildings at least once a year, while interior walls should be cleaned based on the actual situation of the panel cleanliness.
3. The cleaning of curtain walls should be carried out manually or with appropriate cleaning equipment, moving from top to bottom. Avoid using any abrasive materials to scrub the surface. It is recommended to use soft cloths, sponges, or brushes made of non-abrasive materials, and it is best to clean in a consistent direction during the process.
4. Do not clean panel surfaces when their temperature is above 40°C, as rapid water evaporation can affect the metal finish. Always use an appropriate metal cleaning agent and ensure the detergent is neutral. Avoid using strong alkaline detergents (e.g., potassium hydroxide, sodium hydroxide, or sodium carbonate), strong acidic detergents (e.g., nitric acid or sulfuric acid), abrasive cleaners, or cleaners that are corrosive to painted surfaces (e.g., polishing paste, acetone, methyl ethyl ketone, or thinner). Proper cleaning practices help maintain the panel's finish and durability.

5. The specific cleaning steps are as follows:
 - 1) Rinse the panel surface thoroughly with plenty of water.
 - 2) Use a soft brush soaked in diluted detergent to gently scrub the panel surface.
 - 3) Rinse the surface again with plenty of water to remove dirt.
 - 4) Inspect the surface and use detergent to focus on areas that are not fully cleaned.
 - 5) Rinse the surface with clean water until all detergent is completely washed off.
6. Before starting large-scale cleaning, it is recommended to test a small section of the panel to ensure the method is safe and effective.
7. During maintenance, avoid using hard tools for scraping, hammering or striking, and prevent impacts with hard objects to avoid damage on the panel surface.

Examples of commonly used detergents:

- Product name
Turco 5948DPM THICK
- Manufacturer
Henkel Surface Technologies
- Product name
Tegee, TG22
- Manufacturer
Tegee Chemie Bremen GmbH
- Product name
Cream Cleaner K2007
- Manufacturer
Klenco (S) Pte Ltd



FEITENG® DIPPER SERIES

Processing Method

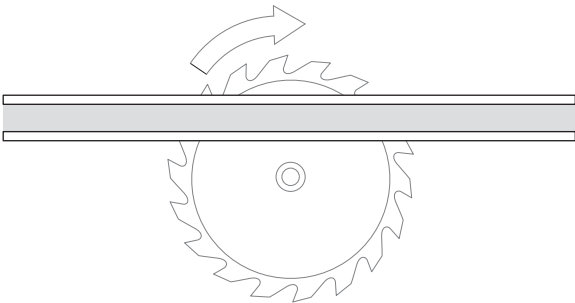


FEITENG® metal composite panels can be easily processed using standard metalworking tools for cutting, punching, grooving and bending. Based on the desired architectural façade design, they can achieve various shapes, including diverse curved surface designs.

Cutting

FEITENG® panels can be cut using common tools such as CNC machines, handheld cutting tools and shearing machines, each suited for different applications: shearing machines are ideal for large-scale production but are limited to simple rectangular cuts, handheld cutting tools are perfect for small-scale processing and on-site modifications, and CNC machines can cut various shapes, including intricate hollow patterns (with hollow sections not exceeding 40% of the panel surface), offering high precision and engraving capabilities.

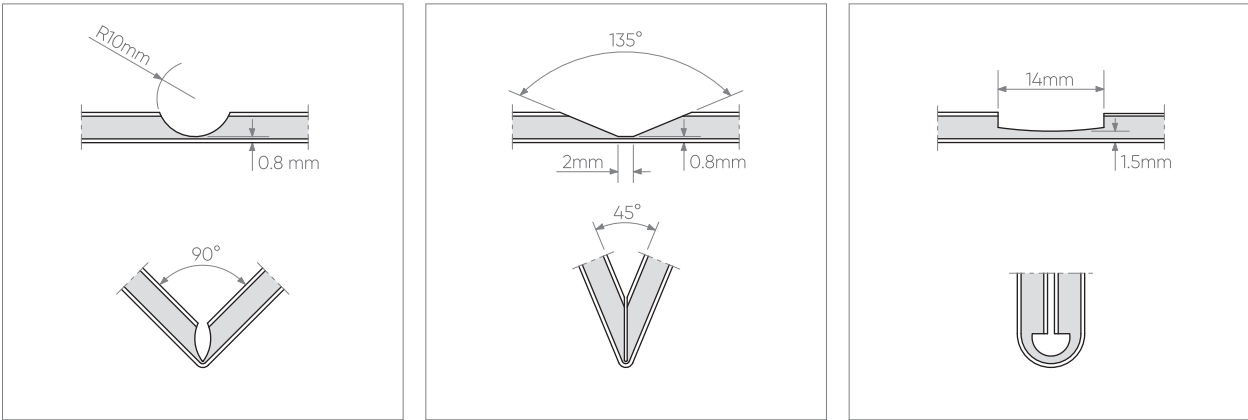
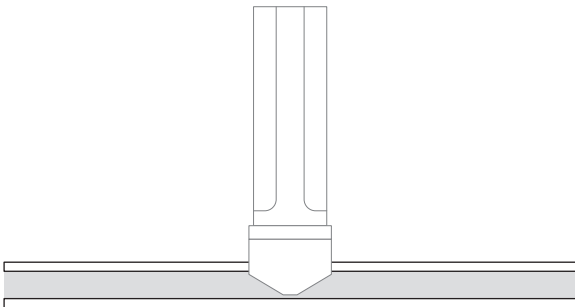
Note: It is important to select the appropriate cutting technique and equipment based on specific requirements, while also adhering to proper operational procedures and regularly maintaining the cutting equipment to ensure optimal performance and cutting quality.



Milling

FEITENG® panels can be milled using CNC machines, BOSCH® trim routers, and BOSCH® handheld circular saws. By changing the corresponding milling tools, different shapes of grooves can be milled into the back of the panel (as shown below). The milling depth should be determined based on the bending requirements.

Note: It is generally recommended to leave at least 0.3mm of core material thickness. The middle core layer should not be entirely milled away, as this would reduce the robustness of the metal surface after bending, potentially shortening the lifespan of the panel.



Spherical milling slot for 90° bends. V-shaped milling slot for 135° bends. Arc milling slot for full 180° bends (not possible on FEITENG® A2 panel).

Recommended Milling Parameters

Composite Panel	Operation	Rotation Speed (R.P.M.)	Displacement Speed (mm/min)
FEITENG® DIPPER SERIES A2	Cutting	22.000-24.000	8.000-10.000
	Milling	24.000	10.000-12.000
FEITENG® DIPPER SERIES FR	Cutting	24.000	12.000-14.000
	Milling	24.000	15.000-18.000

Note: these values are indicative only and are intended for use with FEITENG® recommended cutters. For use with other tools or depending on the type of machine, it is necessary to adjust them.

Recommended tools core A2:

Cutting Tool: Cutter Ø6 mm with diamond inserts (STB-FRESA6PCD)

Milling Tool: Spherical milling machine for fine diamond inserts (STB-FRESA10PCD)

Recommended tools core FR:

Cutting Tool: Cutter Ø6 mm with diamond inserts (STB-FD6X80X3.8)

Milling Tool: Spherical milling cutter for fine diamond coating (STB-FEMD-10x72)

FEITENG® DIPPER SERIES

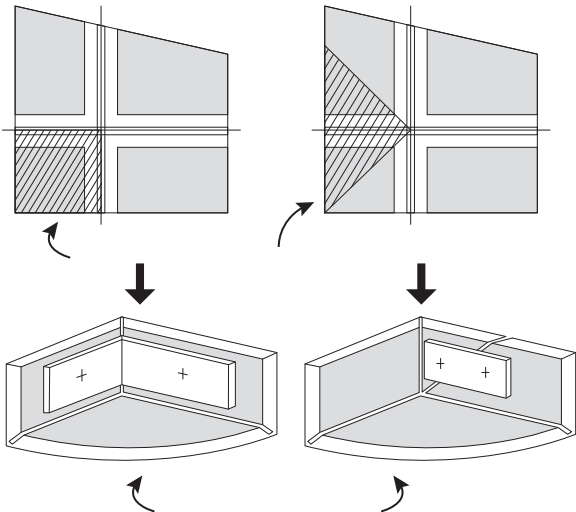
Processing Method



Bending

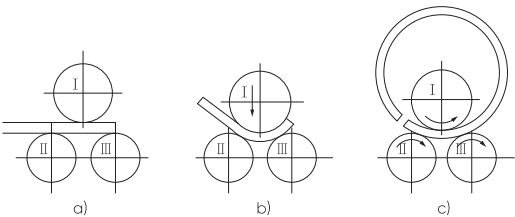
Metal composite panels cannot be directly bent; before bending, a groove must be milled at the bending location. After milling, clean the debris from the groove. When manually bending, use tools to bend the edge in one go, avoiding repeated bending to prevent damage to the surface material. Prior to bending, cut notches around the edges of the panel. After bending, reinforce the areas between the bends using aluminum fittings and rivets, as shown in the diagram below.

Note: The area where the bending tool contacts the metal composite panel should be protected with flexible material. During the bending process, the force should be applied evenly to avoid deformation of the panel.



Coiling Circle

The rolling of metal composite panels requires specialized metal bending equipment, such as bending machines, three-axis bending machines and other bending devices. Before processing, a trial run is needed to determine the number of rolls and the rebound coefficient. During processing, the decorative surface is protected using flexible material between the rollers. Below is the working principle of the symmetric three-roll bending machine:



The suitable bending radius for FEITENG® DIPPER SERIES metal composite panel needs to be derived from the geometric characteristics of the arc and the mechanical principles during the bending process. The following are several common design scenarios:

For simple circular arc bending, the following formula can be used to calculate the bending radius:

$$R = (h^2 + (L/2)^2) / (2h)$$

Among them, R represents the bending radius, L represents the chord length of the arc, and h represents the height of the arc.

1. This formula is suitable for situations where the ratio between the length and height of the arc is moderate. When the length of the arc is much greater than the height, it can be approximated as $R \approx L^2/8h$.

2. For the relationship between the bending radius of an arc and the elastic modulus, bending force, and section moment of inertia of the material, the following formula can be used to calculate:

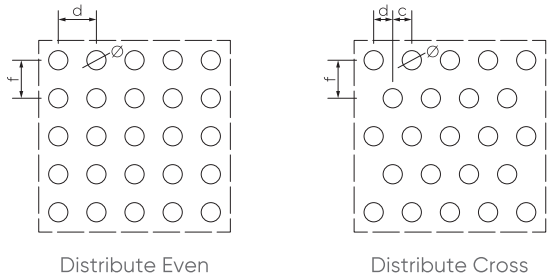
$$R = (E \times I) / M$$

Among them, R represents the bending radius, E represents the elastic modulus of the material, I represents the cross-sectional moment of inertia, and M represents the bending force.

Name	Parameter	Notes
Class A FR	$R \geq 1200\text{mm}$	Requires heating during processing
FR	$R \geq 380\text{mm}$	

Stamping

The surface of FEITENG® metal composite panels can be stamped using specialized CNC machines. The shapes of the punched holes can include round holes, square holes, elongated round holes, hexagonal holes, fish-scale holes, etc. Common arrangements include straight alignment and staggered alignment. Detailed stamping parameters are shown in the diagram below.



Classification	Parameter	Notes
Hole diameter (a)	$\geq 4.5\text{mm}$	
Hole spacing (d, f, e)	$\geq 10\text{mm}$	
Transmittance	$\leq 35\%$	Transmittance = (punching area / total area) x 100%

Riveting

The connection of FEITENG® metal composite panels is done using rivets. Commonly used rivet specifications include M411, M511, although other models can be chosen based on requirements. Rivet connections require holes to be drilled in the composite panel. The distance from the center of the hole to the edge of the panel should not be less than twice the panel thickness. A gap of no more than 0.3mm should be left between the hole and the rivet to avoid compressing and deforming the composite panel.

Adhesion

When the size of the FEITENG® metal composite panel is relatively large, structural reinforcement should be applied to the back of the panel based on the design wind load value of the project. The reinforcement material should be aluminum profiles (which can be U-shaped, C-shaped or square tubes). The aluminum profiles and the back of the panel should be effectively bonded using a two-component silicone structural adhesive (and, if necessary, combined with 3M double-sided adhesive).

Suggest using the following adhesives:

DOW CORNING-DC995
(single component structural adhesive)

SIKA-SG20
(single component structural adhesive)

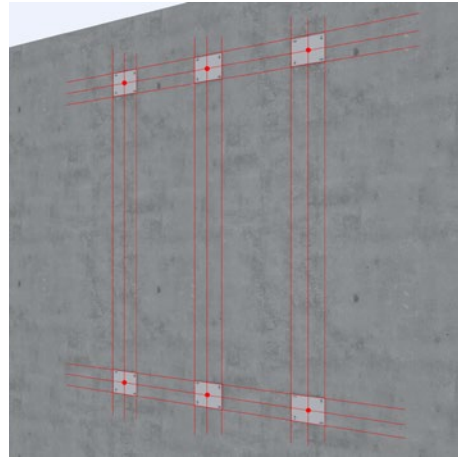
DOW CORNING-DC791
(silicone weather resistant adhesive)

SIKA-WS305CN
(silicone weather resistant adhesive)

DOW CORNING-DC993N
(two component structural adhesive)

Metal Composite Panel Curtain Wall

Installation Steps



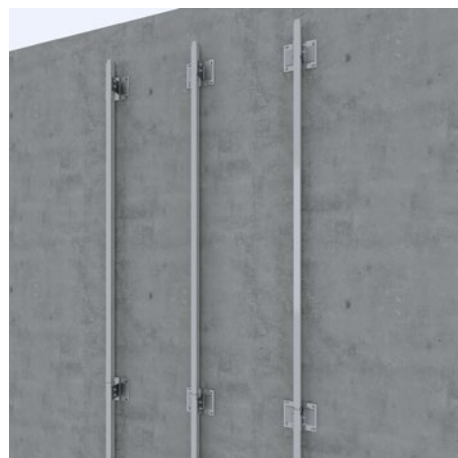
1. Determine the position and size

Based on the curtain wall, construction drawings, architectural drawings, structural drawings, and detailed grid drawings, combined with the elevation control points provided by the civil engineering unit, accurately set out the entry and exit lines and axis positions;



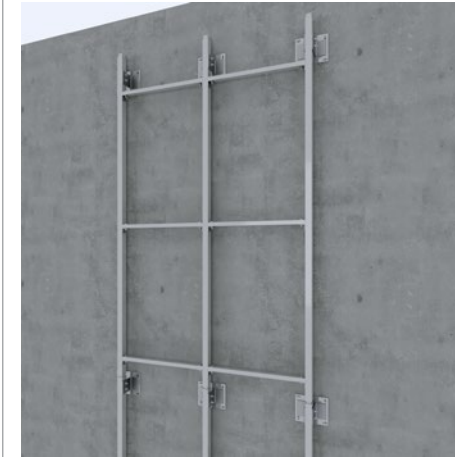
2. Install the embedded parts

Install steel connectors based on the centerline and grid lines marked during the laying out process. The installation sequence is to first spot weld and fix them, check and verify that they meet the requirements, and then fully weld and fix the steel connectors. The height and left and right of the steel connectors should be controlled within 2-3mm;



3. Column installation

The installation of the column should be between the two adapters of the fixed support, and the column should be fixed with stainless steel tension bolts according to the installation elevation requirements. The deviation of the column axis in the front and back and left and right directions should not exceed 2mm and 3mm, respectively;



4. Installation of crossbeam

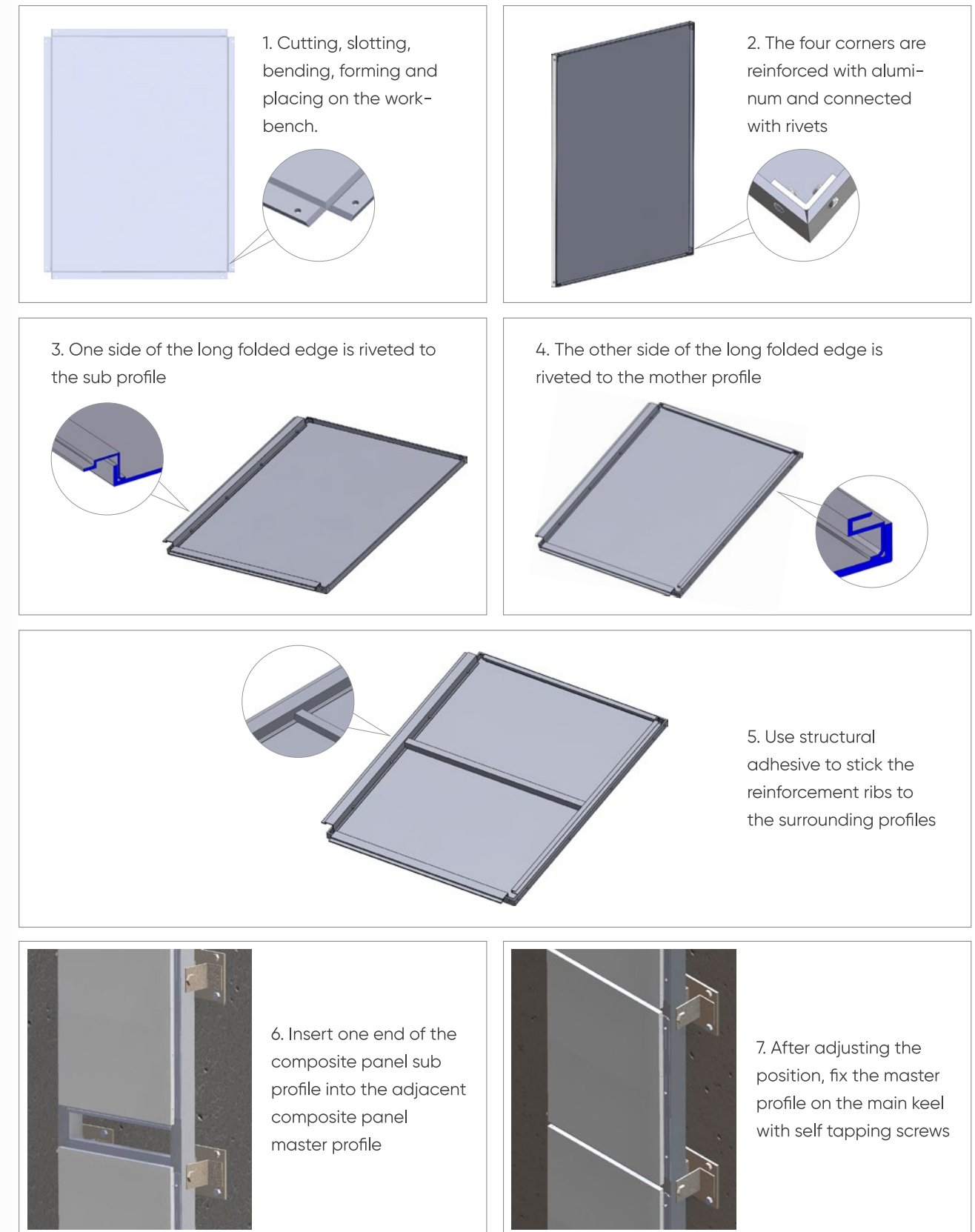
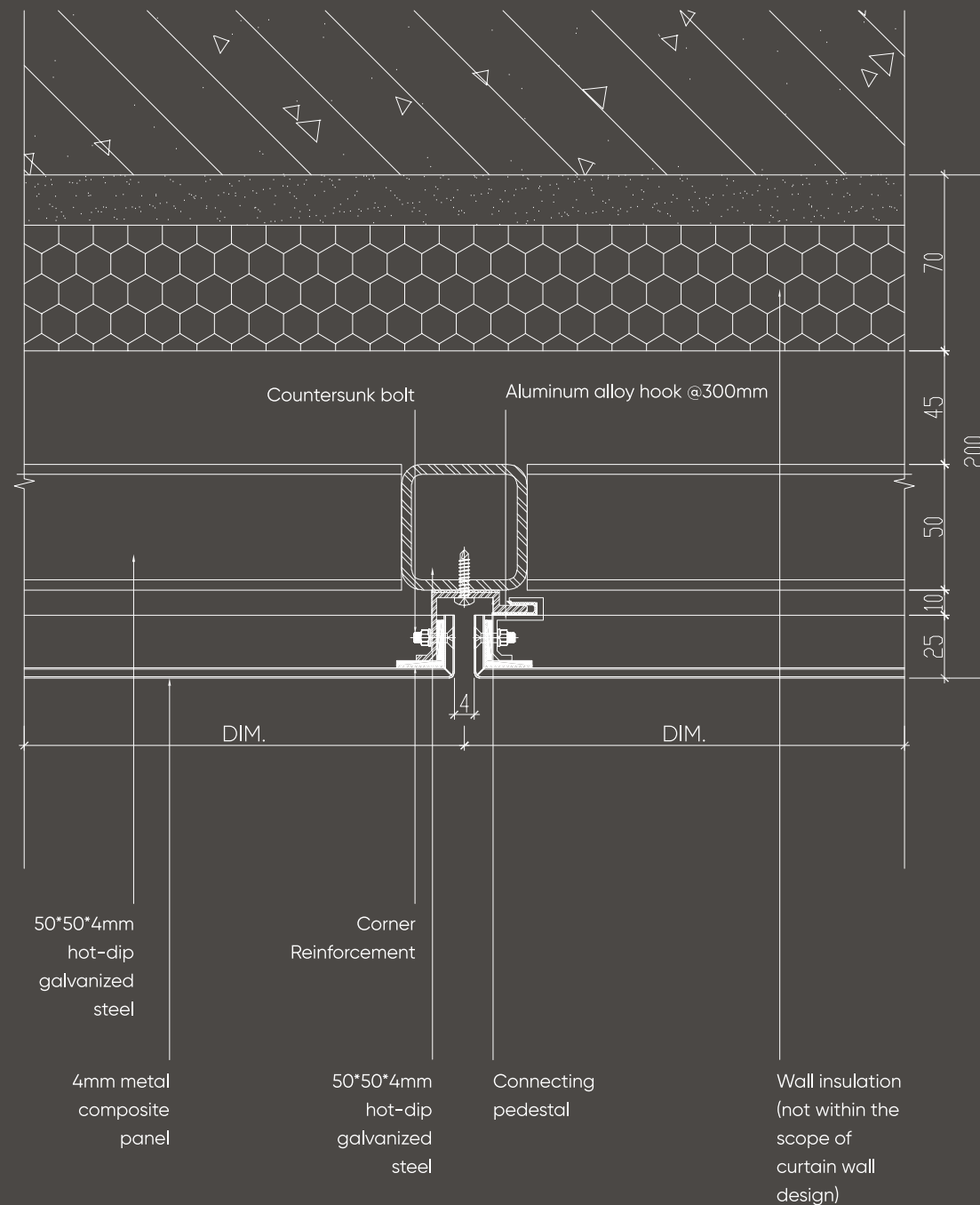
Install the connectors and elastic rubber pads at both ends of the crossbeam at the predetermined position of the column. The installation should be firm and the joints should be tight. The horizontal elevation deviation between adjacent crossbeams should not exceed 1mm. The elevation deviation on the same floor: when the width of a curtain wall is less than or equal to 35m, it should not exceed 5mm; When the width of a curtain wall is greater than 35m, it should not exceed 7mm;



5. Panel installation

Install the metal composite panel layer by layer onto the frame according to the design requirements. During the installation process, attention should be paid to the flatness and verticality of the metal panel, as well as the connection method and sealing treatment with the frame.

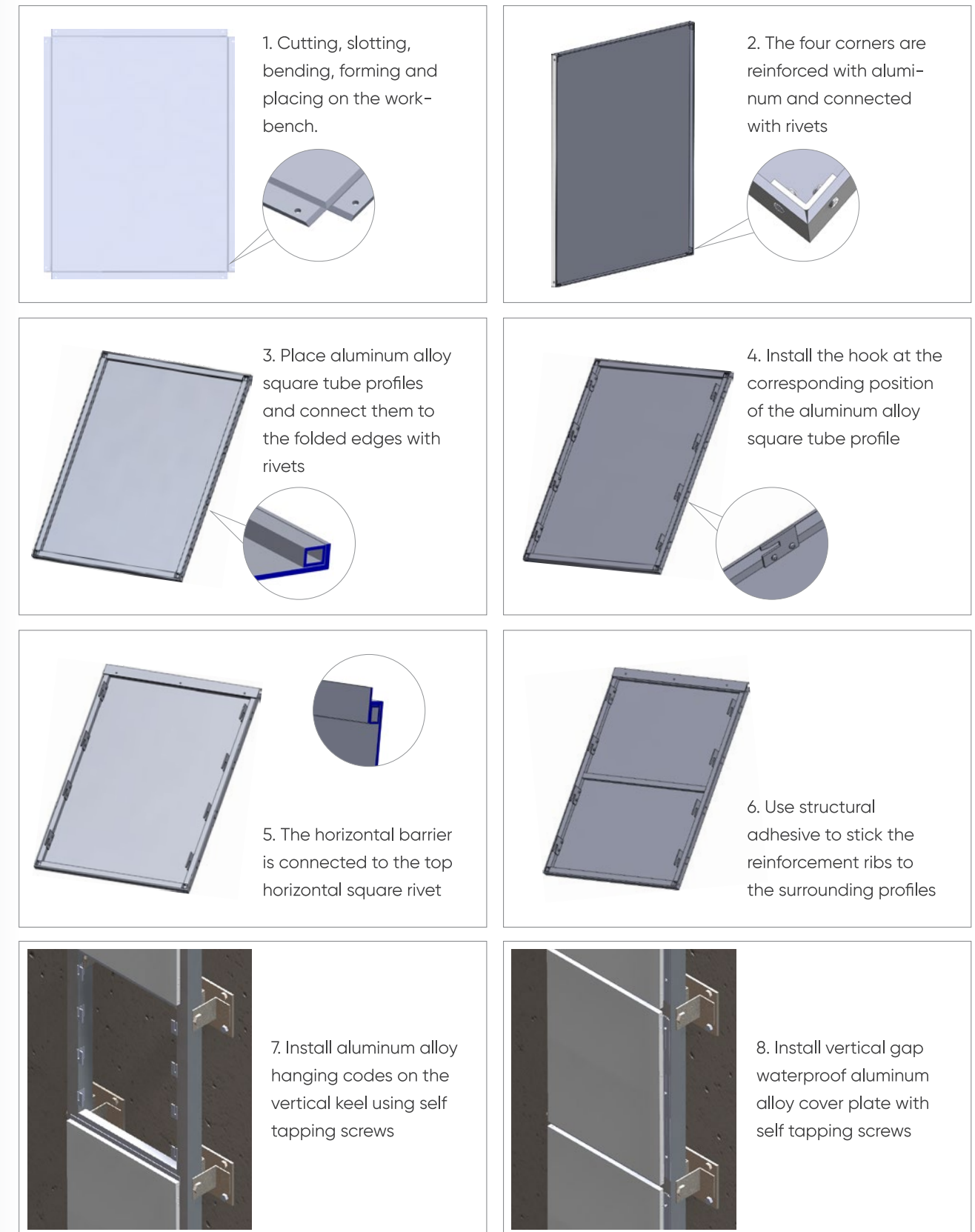
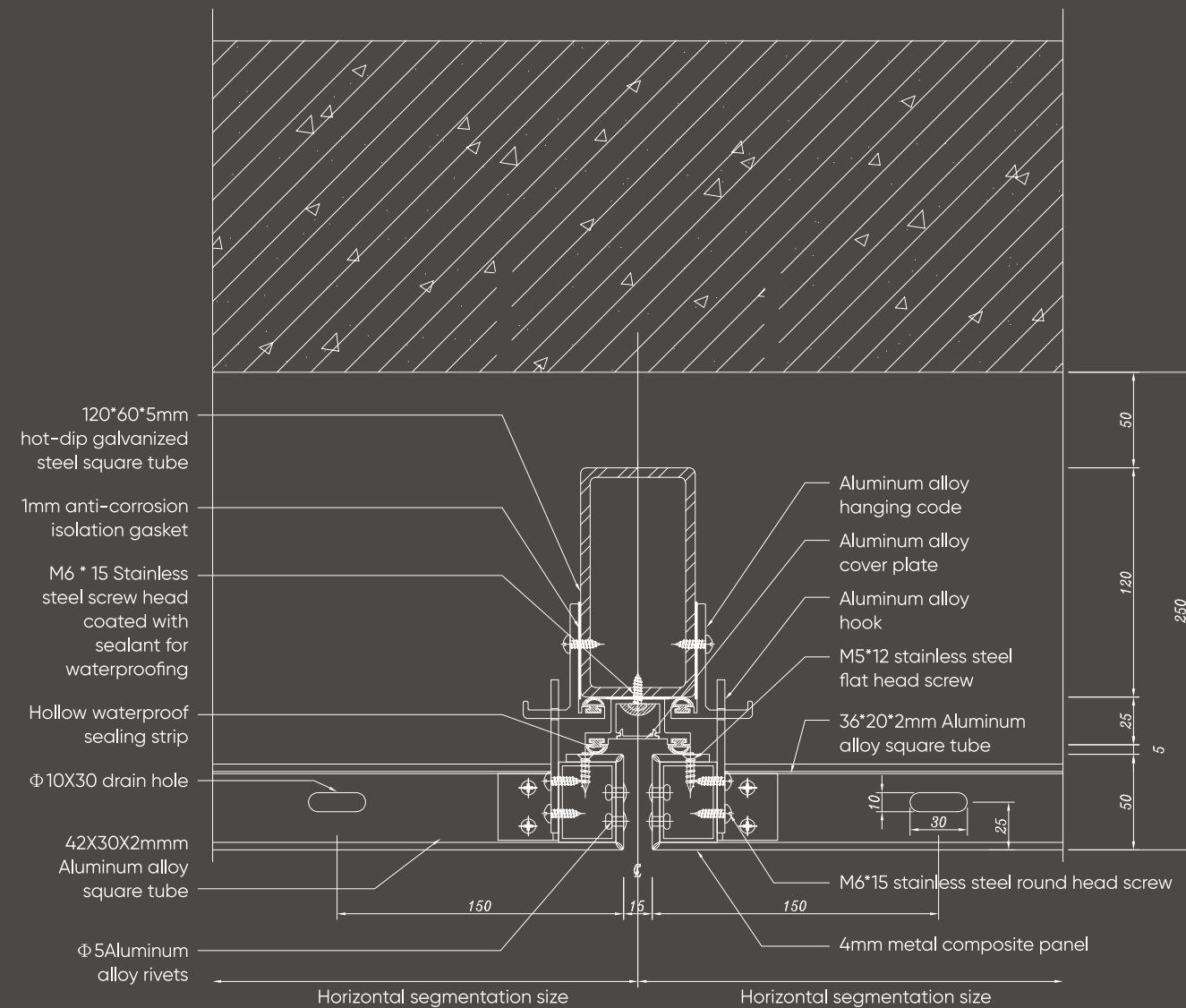
Open Curtain Wall Hanging Code Installation Node



Precautions

1. If M5 rivets are used for riveting holes, it is recommended to open 5.2-5.5 holes;
2. When the panel width is less than 600mm, no reinforcing bars shall be arranged. If the width of the panel exceeds 600mm, the spacing between the reinforcing bars shall be arranged at 500mm, and for panels wider than 1500mm, it shall be determined separately. The reinforcement bars are generally arranged perpendicular to the long side.

Open Curtain Wall Sub Frame Installation Node

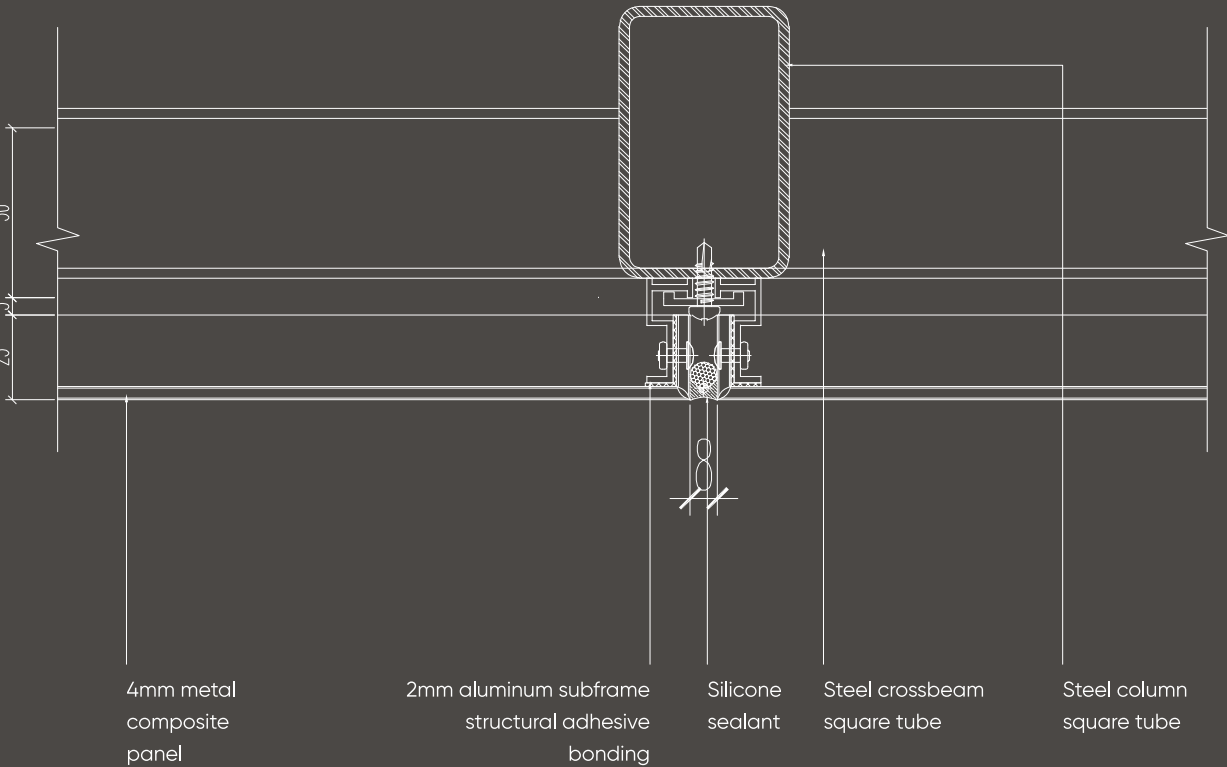


Precautions

1. If M5 rivets are used for riveting holes, it is recommended to open 5.2-5.5 holes;
2. When the panel width is less than 600mm, no reinforcing bars shall be arranged. If the width of the panel exceeds 600mm, the spacing between the reinforcing bars shall be arranged at 500mm, and for panels wider than 1500mm, it shall be determined separately. The reinforcement bars are generally arranged perpendicular to the long side.

Closed Curtain Wall

Sub Frame Installation Node



1. Cutting, slotting, bending, forming and placing on the workbench

2. The four corners are reinforced with aluminum and connected with rivets

3. Place aluminum alloy special profiles and connect them with rivets and folded edges

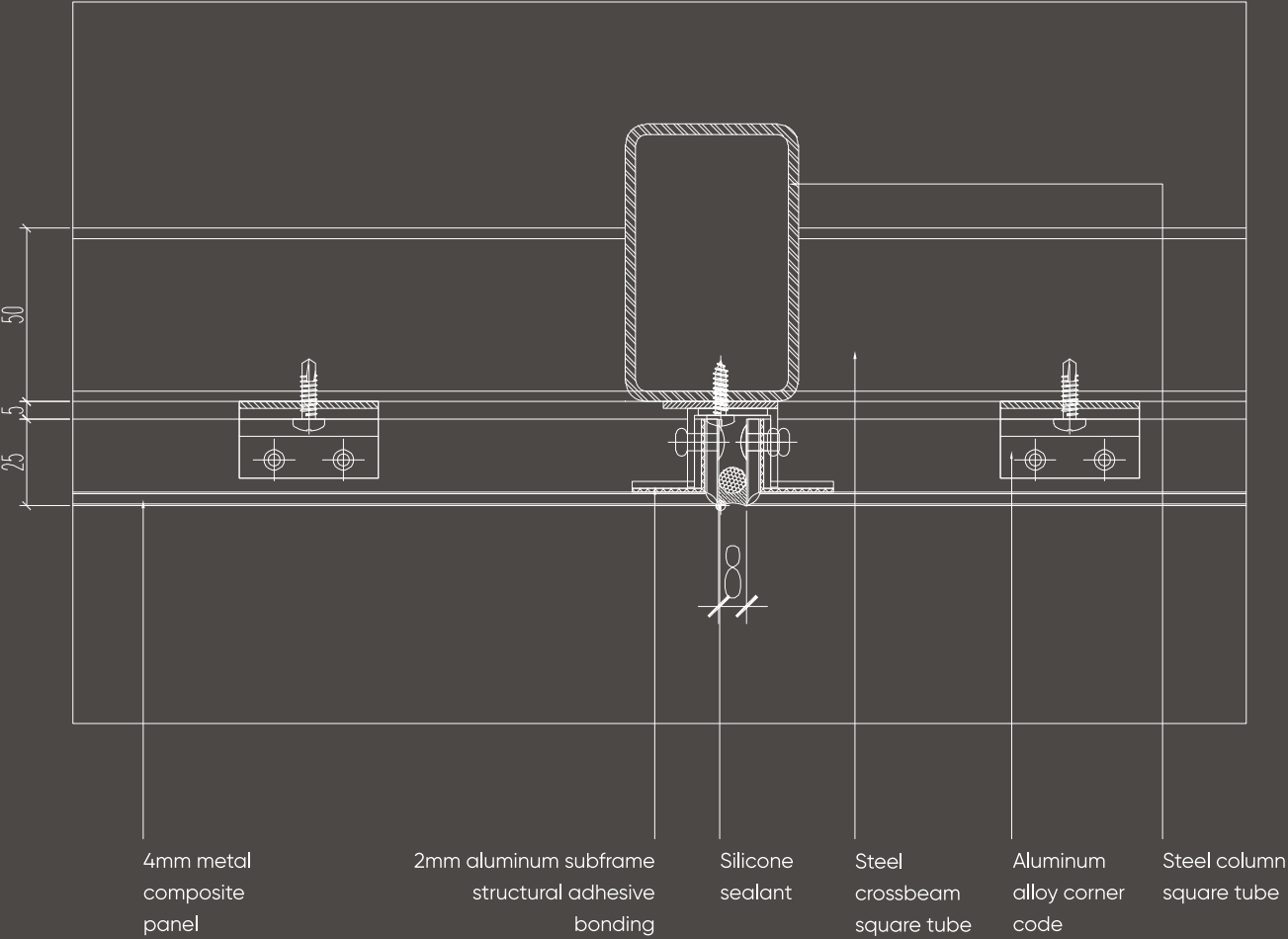
4. Use structural adhesive to stick the reinforcement ribs to the surrounding profiles

5. Insert the block shaped aluminum alloy pressure block through the gap, fix the pressure block with self tapping screws to make the panel profile tightly adhere to the steel keel and leave the design gap

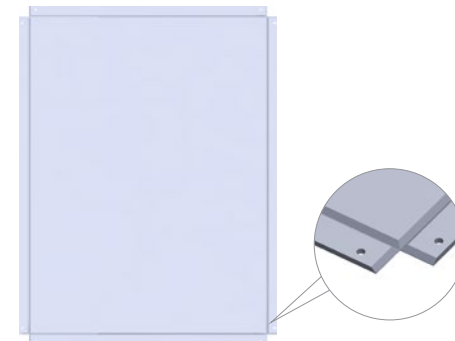
6. Fill the gap with a round foam rod, paste masking paper on the panel, apply sealant, and then tear off the masking paper

Closed Curtain Wall

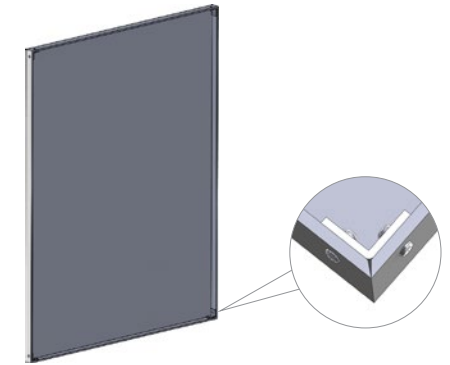
Plug in Installation Node



1. Cutting, slotting, bending, forming and placing on the workbench



2. The four corners are reinforced with aluminum and connected with rivets



3. Place aluminum alloy corner aluminum reinforcement components and use rivets to connect and install corner brackets



4. Use structural adhesive to stick the reinforcement ribs to the surrounding profiles



5. After adjusting the horizontal and vertical positions of the panel, use self tapping screws to fix the corner brackets and leave design gaps.

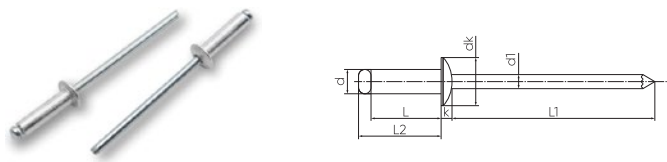


6. Fill the gap with a round foam rod, paste masking paper on the panel, apply sealant, and then tear off the masking paper



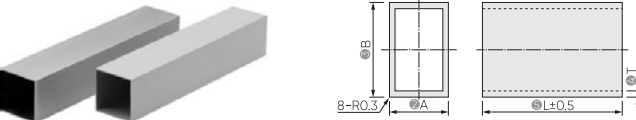
FEITENG®
Accessories

Pull Rivet



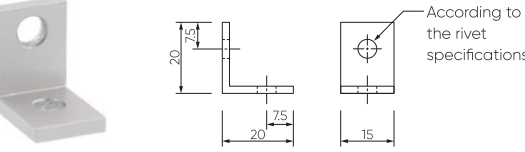
	Specification	3mm	3.2mm	4mm	5mm	6mm
d	Min	2.93	3.11	3.91	4.91	5.91
	Max	3.07	3.29	4.09	5.09	6.09
dk	Min	5.76	5.76	7.71	9.31	11.65
	Max	6.24	6.24	8.29	9.89	12.35
k	Max	1.4	1.4	1.7	2	2.5
d1	≈	1.8	1.8	2.18	2.8	3.6
L2	Max	L+3.4	L+3.5	L+3.5	L+4.0	L+4.5
L1	Min	26	26	27	27	31
L		7-19	7-19	8-20	9-34	10-48

Aluminum Square Pipe

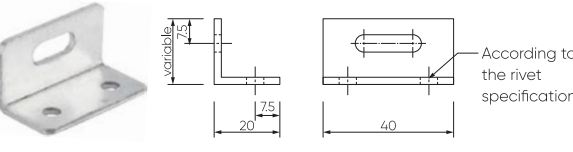


	Specification	①Type	②A	③B	④T	⑤L Designated unit 1mm	Weight kg/m	Cross area mm²	Performance ℓx×10⁴	ℓy×10⁴		
HFHQ	15	15	15	1.5	300-2000	2	0.220	81	0.249	0.249		
							0.342	126	0.454	1.406		
							0.390	144	0.787	0.787		
							0.607	224	1.437	4.445		
							0.497	184	1.634	1.634		
							0.766	284	2.96	9.007		
	20	20	20	2			0.607	224	2.941	2.941		
							0.934	344	5.297	15.94		
							0.824	304	7.336	7.336		
							1.648	464	13.11	38.97		
							1.036	384	14.77	14.77		

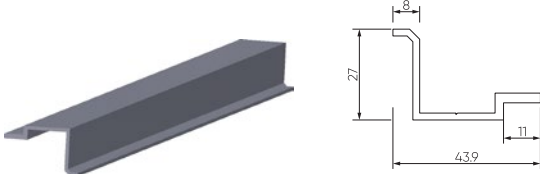
Aluminum Connector



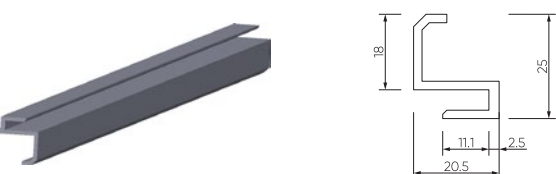
Mounting Angle Code



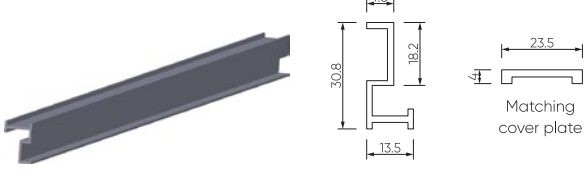
FEITENG Sub Frame 1



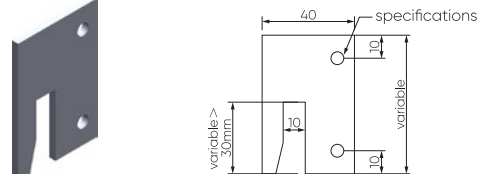
FEITENG Sub Frame 2



FEITENG Sub Frame 3



Hanging Corner Code



31

FEITENG® COMPOSITE PANEL
Cases

Overseas Architectural Curtain Wall Projects

Grand Canyon University (USA)

Product: FR aluminum composite panel
Quantity: 10000m²
Area: Curtain wall
Location: 3300 West Camelback Road, Phoenix, AZ, 85017

32

Arizona State University School of Nursing (USA)

Product: FR aluminum composite panel

Quantity: 10000m²

Area: Curtain wall

Location: Arizona State University, Phoenix, AZ 85004



Nashville Music City Center (USA)

Product: A2 aluminum composite panel

Quantity: 18000m²

Area: Curtain wall

Location: Music City Center, 201 5th Ave S, Nashville, TN 37203

Arizona State University Tempe Campus (USA)

Product: FR aluminum composite panel
Quantity: 5000m²
Area: Curtain wall
Location: Etiwanda Ave, Northridge, CA 91330



Phoenix Airport (USA)

Product: A2 aluminum composite panel
Quantity: 18000m²
Area: Curtain wall
Location: 3400 E Sky Harbor Blvd, FT-111 Silver Metallic Phoenix, AZ 85034

The Poly Pavilion at the University of California, Los Angeles (USA)

Product: A2 aluminum composite panel

Quantity: 10000m²

Area: Curtain wall

Location: 301 Westwood Plaza, Los Angeles, CA 90095



Phoneix Children's Hospital

Product: A2 aluminum composite panel

Quantity: 12000m²

Area: Curtain wall

Location: 1919 E thomas Rd F12, Phoneix, AZ 85016



Arizona State University School

Product: A2 aluminum composite panel
Quantity: 8000m²
Area: Curtain wall
Location: 411 N Central Ave, Phoenix, AZ 85004



Gateway Community University (USA)

Product: A2 aluminum composite panel
Quantity: 16000m²
Area: Curtain wall
Location: 108 N 40th St. Phoenix. AZ 85034

Scottsdale Quarter – Block M (USA)

Product: FR aluminum composite panel

Quantity: 8000m²

Area: Curtain wall

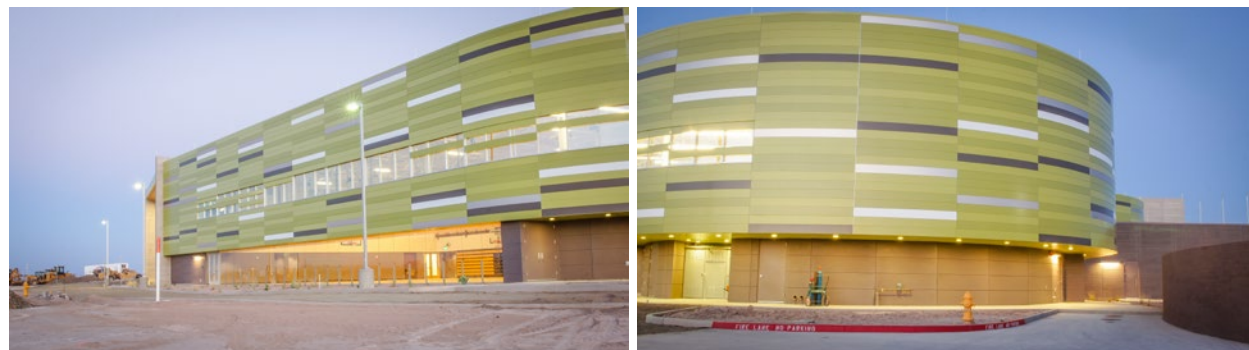


University of Sydney International Student Apartments (Australia)

Product: A2 aluminum composite panel

Quantity: 8000m²

Area: 9 Kensington St, Chippendale NSW 2008



Gila River MP Building (USA)

Product: A2 aluminum composite panel

Quantity: 20000m²

Area: Curtain wall

Location: 8070 Park St, Sacaton, 85147

China Aid to Africa Railway Tazara Memorial Hall

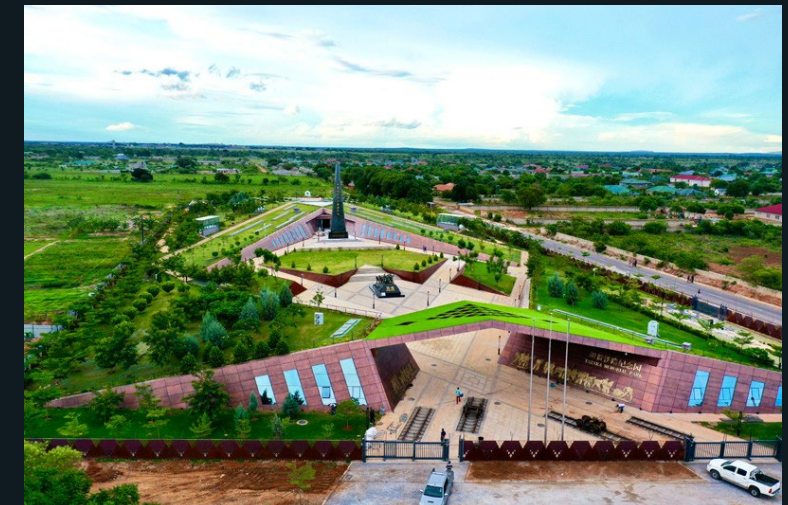
Product: FR copper composite panel /

Copper & aluminum composite panel

Quantity: 5000m²

Area: Curtain wall

Location: Plot No. LN/4034/1 Along Great East Road in Silverest Area of Chongwe District





Chandler City Hall

Product: FR aluminum composite panel
Quantity: 8000m²
Area: Curtain wall
Location: 175 S Arizona Ave, Chandler, AZ 85225

FEITENG® COMPOSITE PANEL Cases

China Architectural Curtain Wall Projects

Haikou Bay Performance Center

Product: A2 aluminum composite panel
Quantity: 11000m²
Area: Curtain wall
Location: No. 42 Binhai Avenue, Jinmao Street, Longhua District, Haikou City, Hainan Province



Huanglong Vanke Center

Product: A2 aluminum composite panel
Quantity: 10000m²
Area: Curtain wall
Location: No.8 Wen'er Road, Xihu District, Hangzhou City, Zhejiang Province



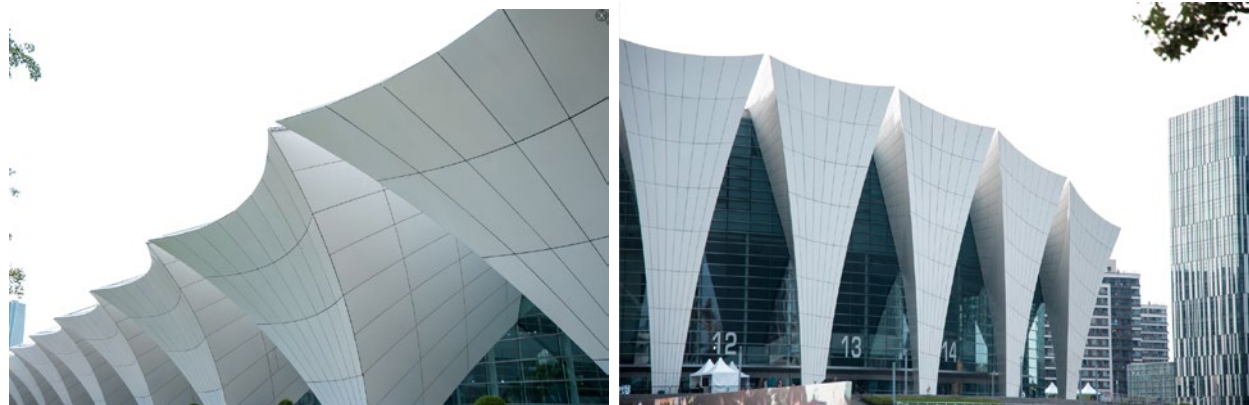
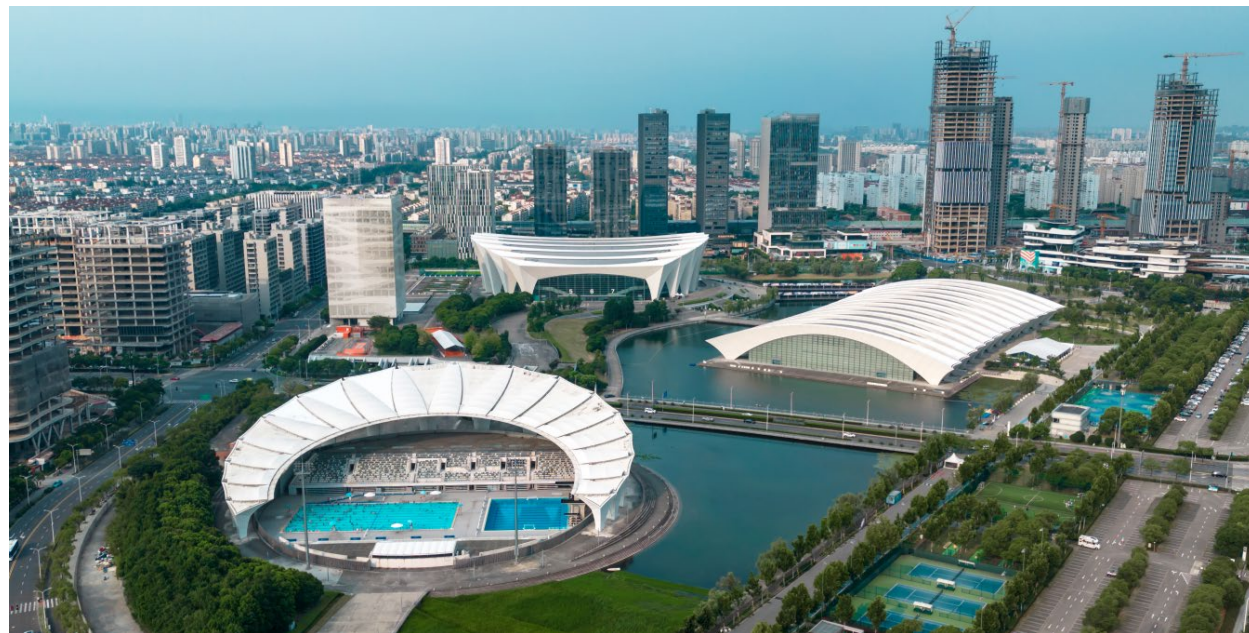
Shanghai Oriental Sports Center

Product: A2 aluminum composite panel

Quantity: 12000m²

Area: Curtain wall

Location: 5F3J+WCB, Yaoti Road, Shanghai



Vancreate Industrial Park

Product: FR aluminum composite panel

Quantity: 12000m²

Area: Perforated curtain wall

Location: No.20 Chengyang Road, Zhangjiagang City, Jiangsu Province



Zhangjiagang Vancreate Industrial Park Cultural Center



JinJiang International Yantai
Fontoo Hotel



Shanghai Lingang Teenagers
Activity Center



DNE Group Intelligent Manufacturing
Park Nantong Base



The Fourth Affiliated Hospital of Nanjing Medical University



Zhangjiagang Funworld



Shanghai Bright Intelligent
Curtain Wall



Shanghai Guoqing Temple



Leshan Olympic Center



Shanghai CMOS Image Chip Project



Shanghai Jing'an China Tower



Sam's Club Wuxi Branch



Shanghai Sany Heavy
Machinery Co., Ltd



Shanghai Tobacco Group



Shanghai China Financial Building



Shanghai Port International Shipping Center



Hefei Government Affairs Culture New District Junior High School



Hefei City Huating



Aerospace Third and Fourth Hospital



Tianjin Junyi Building



Tsinghua Tongfang Office Building



Shanghai Greenland Building



Yantai International Airport



Kunming Yinde Building



Tianjin Xiali Building



Chongqing High-tech Park Mercury Technology Building



Chongqing High-tech Park Venus Technology Incubation Center



Hohhot Railway Station



Theatre of Changchun Hydrologic Culture Garden



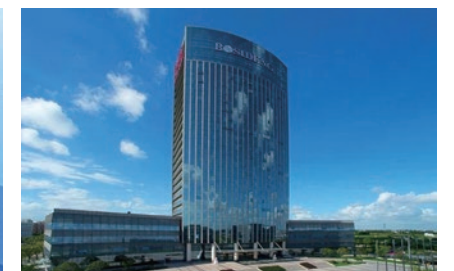
Zhangjiagang Agricultural Bank



Shanghai Metropolis



Chongqing Jiangbei Airport



Bosideng Global Headquarters Building



Xi'an Aircraft Factory



Beijing Institute of Nuclear Industry



New Nanjing Central Shopping Mall



Haikou Huafu Building